

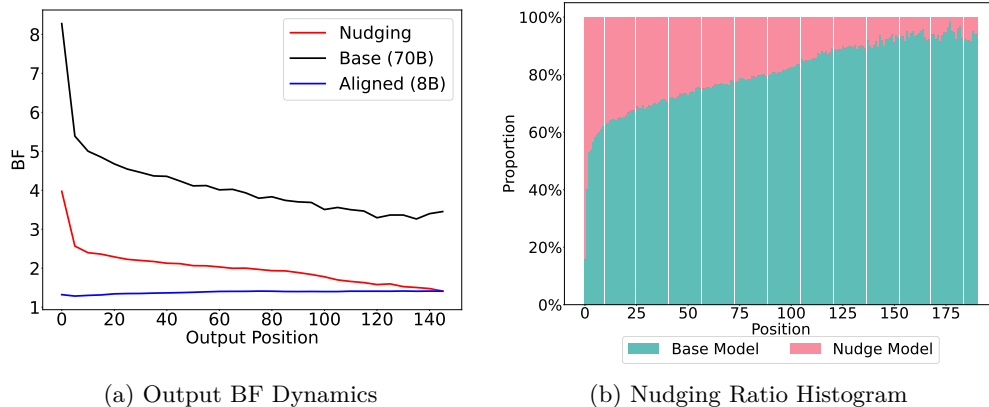


Figure 6: Nudging Experiments over Just-Eval-Instruct.

this view, alignment tuning doesn’t reshape generation from scratch, but instead nudge the model toward *stylistic tokens* (e.g., “Sure”), thereby narrowing the conditional distribution.

To test this hypothesis, we reproduce the nudging experiments (Fei et al., 2024), over Just-Eval-Instruct (Lin et al., 2023) and MMLU datasets. We employ Llama-3-70B for drafting most outputs. However, when the base model’s Top-1 probability is low, we apply nudging by switching to Llama-3-8B-Instruct to generate a single word. Using a smaller aligned model to nudge a larger base model (70B) isolates the steering effect of the stylistic prefix itself, independent of the nudging model’s raw capability. BF was computed as in prior experiments. The results, shown in Figure 6,⁹ indicate that after most nudging occurs early in the generation process – indicating the prefix generated by the nudging model is of low probability. These observations collectively support our hypothesis. Considering that nudging not only reduces BF but also improves aligned model performance on these tasks (Fei et al., 2024), our results highlight the dual effect of alignment training: reducing BF while preserving or even enhancing task performance.

Which Training Stage Reduces BF Most? While our nudging analysis suggests alignment tuning narrows the distribution by nudging models toward stylistic tokens, it remains unclear which specific phase of the pipeline drives this reduction. Disentangling these effects is difficult because most model releases (e.g., Llama 3 (Dubey et al., 2024)) do not provide intermediate checkpoints, and modern post-training often employs iterative, interleaved schedules rather than discrete stages. However, the OLMo-2 suite (OLMo et al., 2024) releases intermediate checkpoints, enabling a preliminary stage-wise dissection. We measure BF changes across Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) (Rafailov et al., 2023) stages on the open-ended CREATIVE STORYGEN task. As shown in Figure 7, we observe distinct behaviors across model scales. For OLMo-2-7B, SFT is the primary driver of BF reduction (causing a 42.3% drop vs. 20.5% for DPO). In contrast, for OLMo-2-13B, the DPO stage contributes more significantly (25.6% drop vs. 14.9% for SFT). This divergence suggests that the impact of alignment tuning stages is not universal but sensitive to specific training recipes and model scales. Future work with fine-grained checkpoint trajectories is needed to fully map these dynamics.

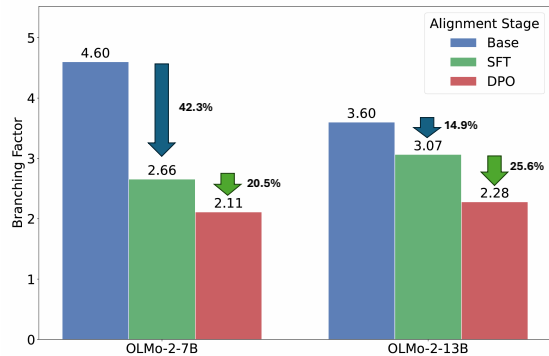


Figure 7: **Stage-wise Contribution to BF Reduction.** We analyze OLMo-2 models on the Creative StoryGen task.

⁹We present results on Just-Eval-Instruct only for brevity. MMLU results are included in § I.

7 Related Works

Uncertainty Quantification for LLM. Uncertainty quantification (UQ) for LLMs has gained significant attention due to its importance in real-world applications, particularly in high-stakes domains (Desai & Durrett, 2020; Jiang et al., 2021; Wang et al., 2022; Kadavath et al., 2022; Xiong et al., 2024; Ye et al., 2024; Gupta et al., 2024). Existing methods typically address closed-domain tasks such as classification and question-answering, where outputs are discrete and easier to assess. However, as Kuhn et al. (2023) note, these approaches often overlook challenges specific to open-ended generation, such as semantic equivalence across outputs. They introduce “semantic entropy” to quantify uncertainty in LLM output space by first clustering the sampled output and then quantifying uncertainty over cluster distribution. This method empirically works well in hallucination detection (Farquhar et al., 2024). In this paper, we focus on investigating the probability concentration phenomenon for LLMs. We introduce BF to quantify this concentration, which applies broadly across tasks without imposing strong assumptions on output categories.

Reduced Diversity in Aligned Models. Recent studies have consistently shown that alignment tuning reduces output diversity in language models (Perez et al., 2022; Padmakumar & He, 2024; Chakrabarty et al., 2024; Tian et al., 2024; Kirk et al., 2024; Lu et al., 2025; Lake et al., 2025; West & Potts, 2025). Mechanistically, Lake et al. (2025) observe that alignment suppresses diversity by aggregating information into longer, standardized responses, though they argue this preserves useful base model behaviors. Our work builds on this inquiry by connecting reduced diversity with related observations on diminished randomness and robustness in aligned models (Saparov & He, 2023; Song et al., 2024; Renze & Guven, 2024; Bigelow et al., 2024), and proposes a unifying explanation: increased probability concentration. Traditional diversity metrics such as n-gram lexical diversity (Li et al., 2016) are sensitive to vocabulary size and output length (Liu et al., 2022; Tevet & Berant, 2021; Guo et al., 2024; Kirk et al., 2024) and cannot work well with most recent long CoT models. In § K, we demonstrate that lexical diversity poorly correlates with BF and fails to robustly measure generation concentration.

Our work also resonates with information density research in cognitive science and linguistic theories, and we present a short discussion in § J.

8 Discussion

Practical Implications. A key practical implication of our findings is that reduced BF neglects alternative generations and forking. Consequently, simply tweaking decoding parameters (e.g., temperature), is unlikely to restore diversity without severely degrading quality (Renze & Guven, 2024). Our work offers a clear explanation for why this occurs, particularly for methods like beam search. The resampling experiment in § 5.3 provides direct evidence that for low-BF models, off-path trajectories are not just less probable but often of lower quality. With little probability mass distributed among alternative paths, beam search has few viable options to explore, yielding diminishing returns. This suggests that efforts to mitigate diversity loss should target the training process itself – a more promising, albeit challenging, direction. Future work could involve curating more diverse alignment data or designing novel training objectives that balance instruction-following with distributional diversity (Wang et al., 2024; Kwon et al., 2024; Lanchantin et al., 2025; Chung et al., 2025). System-level interventions (e.g., model collaboration) also present a viable path forward (Fei et al., 2024; Lu et al., 2024; Venkatraman et al., 2025; Ismayilzada et al., 2025). While our paper’s primary contribution is diagnostic, we believe this foundational understanding is a necessary prerequisite for developing such effective countermeasures.

Societal Homogeneity Bias of Alignment Tuning. Our work identifies a key dynamic in modern LLMs: alignment tuning significantly reduces the Branching Factor (BF), leading to more homogenized and predictable outputs. While this can be beneficial, it also carries potential negative societal impacts. In applications such as automated content generation, creative writing, or decision-support systems (Padmakumar & He, 2024; Sorensen et al., 2024; Wu et al., 2025a; Murthy et al., 2025; Rodemann et al., 2025; Ashkinaze et al., 2025; Lake et al., 2025), this reduction in diversity could inadvertently reinforce social biases, stifle creativity, and limit the exploration of novel ideas. Rodemann et al. (2025) further argue that empirical alignment, relying on limited and potentially biased human feedback, creates selectional bias that fails to

capture the full spectrum of human values. We believe that formally understanding and quantifying the mechanisms of probability concentration, as we do in this paper, is a critical and necessary first step toward developing alignment techniques that mitigate these risks and foster models that are not only helpful and harmless but also diverse and robust.

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